

COURSE: AI Assisted Coding

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BATCH-34

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Lab 3: Prompt Engineering: Improving prompt and context management

Objective: To explore how variations in prompt structure affect the quality, completeness, and accuracy of responses from a large language model.

Requirements:

- VS Code with GitHub Copilot or Cursor API and/or Google Colab with Gemini
- Tasks to be completed are as below.

Task 1: Conceptual Understanding in Physics

Scenario

Suppose that you are a data assistant developer for an EdTech company that uses ChatGPT to answer student queries related to introductory physics.

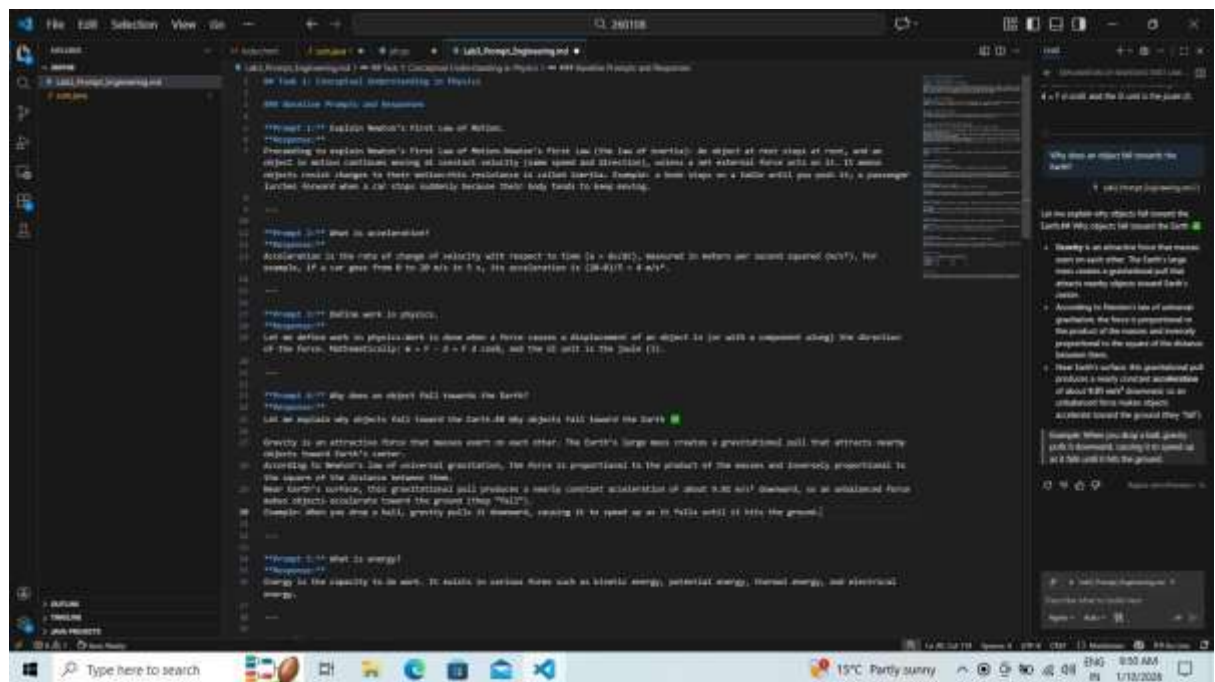
Tasks to be completed

1. Baseline Prompt Testing

Choose 5 typical user queries, for example:

“Explain, What, Define, Why, What”

- Run these prompts in a chat-based AI model and record the raw responses.

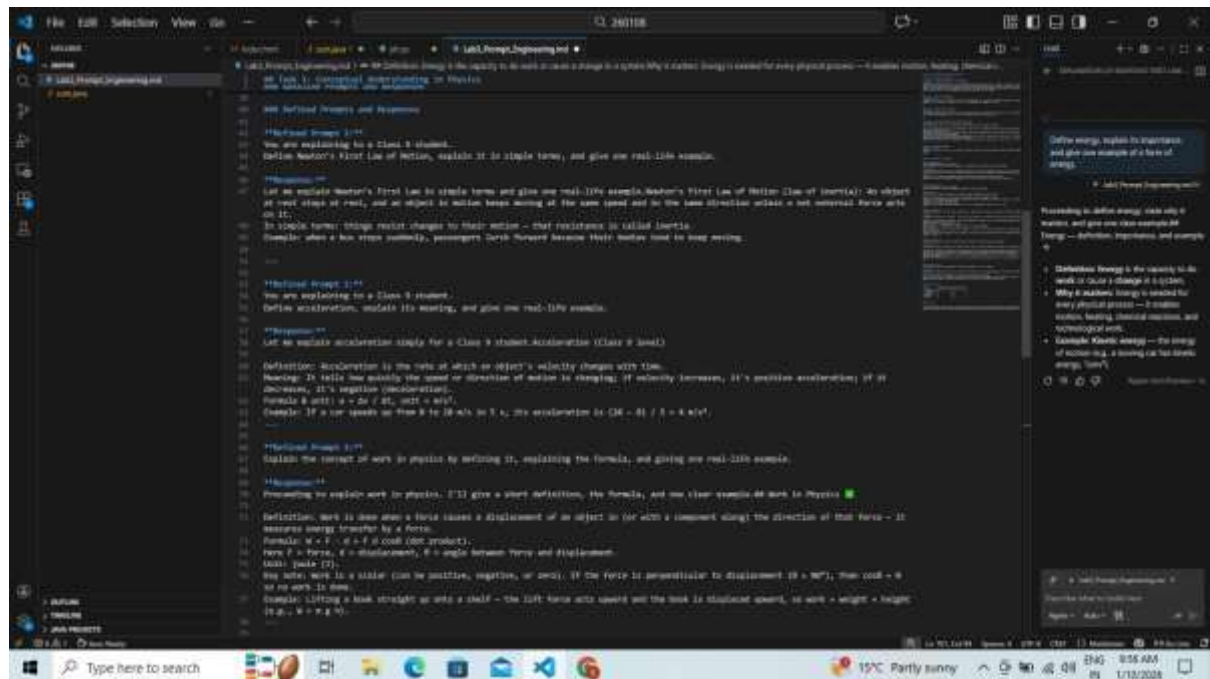


2. Prompt Refinement

Rewrite each query using the following strategies:

- Add 5 different contexts (school level, competitive exam, real-life application, mathematical focus, conceptual focus).
- Make the task explicit (e.g., “List and explain Newton’s three laws with one real-world example each.”).
- Break the query into subtasks (definition → explanation → example).

Run these prompts in a chat-based AI model and record the raw responses.



3. Evaluate Outputs

Score AI responses on a scale of 1–5 using:

- Completeness
- Accuracy
- Relevance
- Clarity

Run these prompts in a chat-based AI model and record the raw responses.

Present results in a comparative table.

4. Reflection

Discuss how contextual and structured prompts influenced the depth and correctness of responses.



Task 2: Programming Fundamentals (Python)

Scenario

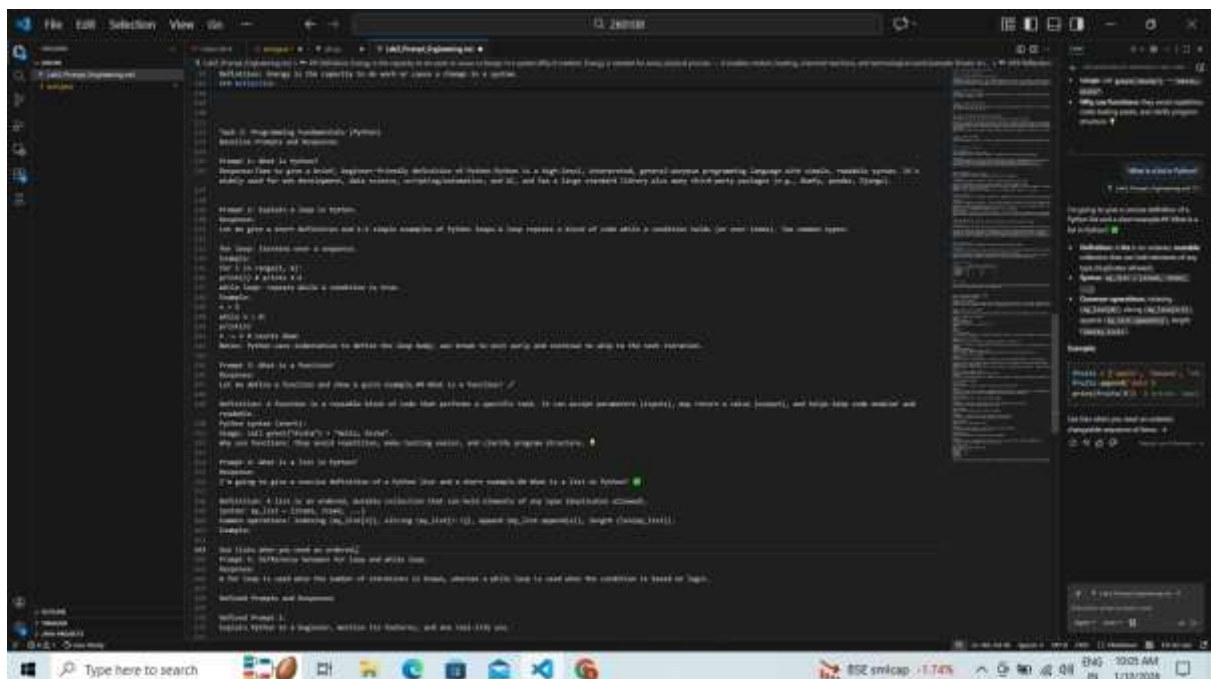
Suppose that you are a data assistant developer for an EdTech platform that supports beginner programming students. Tasks to be completed

1. Baseline Prompt Testing

Choose 5 common user queries, such as:

- “What, Explain, How, What, Difference”

Run these prompts in a chat-based AI model and record the raw responses.



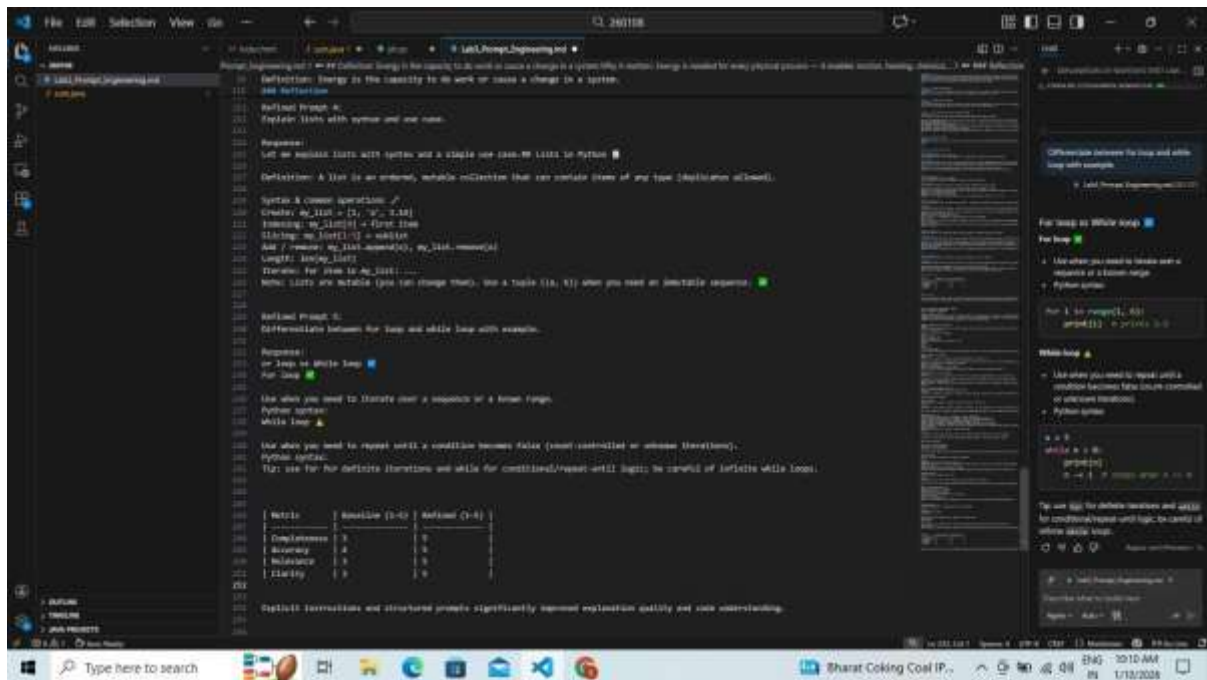
2. Prompt Refinement Rewrite each query by:

- Adding 5 contexts (beginner, exam-oriented, real-world analogy, syntax-focused, performance-focused).
- Making instructions explicit (e.g., “Define a Python loop and show one example for forloop and while-loop.”).
- Breaking into subtasks (definition → syntax → example → use case).

Run these prompts in a chat-based AI model and record the raw responses.

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1 # task_prompts_beginning.py
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3 Prompt: beginning and I want definition: being is the capacity to do with or cause a change in a system (often a matter). Being is needed to study physical process — it enables motion, feeling, thinking... I want 400 characters
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Task 3: Data Science and Machine Learning Concepts

Scenario

Suppose that you are a data assistant developer for an EdTech company offering data science courses.

Tasks to be completed

1. Baseline Prompt Testing

Select 5 typical queries, for example:

- “What, Explain, What, Define, What”

Run these prompts in a chat-based AI model and record the raw responses.



2. Prompt Refinement Refine each query by:

- Adding 5 contexts (academic, industry, beginner-friendly, mathematical, interview-focused).
- Making tasks explicit (e.g., “Define supervised learning and explain it with one real-world example.”).
- Breaking into subtasks (definition → types → example → limitation).

Run these prompts in a chat-based AI model and record the raw responses.

3. Evaluate Outputs

Score outputs using the given metrics and present results in a table.

4. Reflection

Reflect on how context management affects conceptual clarity in technical domains.

Task 4: Database and SQL Queries

Scenario

Suppose that you are a data assistant developer supporting students learning database systems.

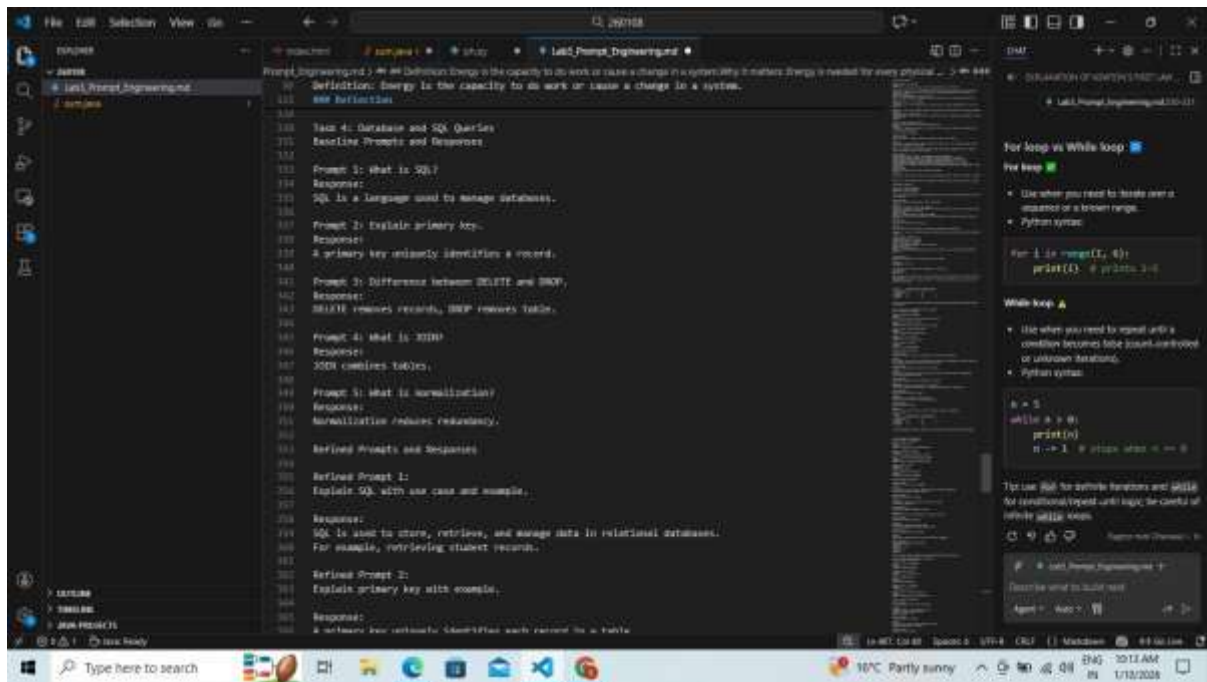
Tasks to be completed

1. Baseline Prompt Testing

Choose 5 common queries, such as:

- “Explain, What, Difference, where, how”

Run these prompts in a chat-based AI model and record the raw responses.



2. Prompt Refinement Rewrite each prompt by:

- Adding 5 contexts (theory exam, practical lab, interview prep, real-world database, optimization focus).
- Making instructions explicit (e.g., “Explain SQL JOIN types with syntax and examples.”).
- Breaking into subtasks (definition → syntax → example → use case).

Run these prompts in a chat-based AI model and record the raw responses.

3. Evaluate Outputs

Evaluate responses using the four metrics and summarize results in a comparison table.

4. Reflection

Discuss how refined prompts reduce ambiguity in technical explanations.

- Adding 5 contexts (school exams, competitive exams, real-life analogy, formula-based, step-by-step solving).
- Making tasks explicit (e.g., “Define probability and solve one simple numerical example.”).
- Breaking into subtasks (definition → formula → example → common mistakes).

Run these prompts in a chat-based AI model and record the raw responses.

3. Evaluate Outputs

- Score responses using completeness, accuracy, relevance, and clarity.
- Present findings in a table.

4. Reflection

Reflect on how structured prompts improve step-by-step reasoning and learner understanding.

The screenshot shows a VS Code editor with a Python script named `task_prompt_engineering.py` and a Jupyter Notebook named `task_prompt_engineering.ipynb`.

Python Script Content:

```

1 # Definition: Energy is the capacity to do work or cause a change in a system (like a battery). Energy is needed for many physical processes - it enables motion, heating, etc.
2 # Task: Define energy.
3
4 # Prompt 1:
5 Define probability with formula and example.
6
7 # Response:
8 Probability measures likelihood of an event.
9 Formula:  $\text{Probability} = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}}$ 
10 Example: Tossing a coin.
11
12 # Prompt 2:
13 Explain average with example.
14
15 # Response:
16 Average is total value divided by number of values.
17 Example: Average marks of students.
18
19 # Prompt 3:
20 Explain ratio with example.
21
22 # Response:
23 Ratio compares quantities.
24 Example: Boys to girls ratio.
25
26 # Prompt 4:
27 Explain percentage with real-life use.
28
29 # Response:
30 Percentage represents value per hundred.
31 Example: Exam scores.
32
33 # Prompt 5:
34 Explain logical reasoning with example.
35
36 # Response:
37 Logical reasoning involves step-by-step thinking.
38 Example: Solving puzzles.
39
40 # Evaluation Table
41 # Metrics: Completeness (1-5), Accuracy (1-5), Relevance (1-5), Clarity (1-5)
42
43 # Prompt 1
44 | Metric | Score |
45 |---|---|
46 | Completeness | 4 |
47 | Accuracy | 5 |
48 | Relevance | 5 |
49 | Clarity | 5 |
50
51 # Prompt 2
52 | Metric | Score |
53 |---|---|
54 | Completeness | 4 |
55 | Accuracy | 5 |
56 | Relevance | 5 |
57 | Clarity | 5 |
58
59 # Prompt 3
60 | Metric | Score |
61 |---|---|
62 | Completeness | 4 |
63 | Accuracy | 5 |
64 | Relevance | 5 |
65 | Clarity | 5 |
66
67 # Prompt 4
68 | Metric | Score |
69 |---|---|
70 | Completeness | 4 |
71 | Accuracy | 5 |
72 | Relevance | 5 |
73 | Clarity | 5 |
74
75 # Prompt 5
76 | Metric | Score |
77 |---|---|
78 | Completeness | 4 |
79 | Accuracy | 5 |
80 | Relevance | 5 |
81 | Clarity | 5 |
82
83 # Reflection
84 Structured prompts improved step-by-step reasoning.

```

Jupyter Notebook Content:

Definition: Energy is the capacity to do work or cause a change in a system (like a battery). Energy is needed for many physical processes - it enables motion, heating, etc.

Task: Define energy.

Prompt 1: Define probability with formula and example.

Response: Probability measures likelihood of an event. Formula: $\text{Probability} = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}}$ Example: Tossing a coin.

Prompt 2: Explain average with example.

Response: Average is total value divided by number of values. Example: Average marks of students.

Prompt 3: Explain ratio with example.

Response: Ratio compares quantities. Example: Boys to girls ratio.

Prompt 4: Explain percentage with real-life use.

Response: Percentage represents value per hundred. Example: Exam scores.

Prompt 5: Explain logical reasoning with example.

Response: Logical reasoning involves step-by-step thinking. Example: Solving puzzles.

Evaluation Table

Metric	Score (1-5)
Completeness	4
Accuracy	5
Relevance	5
Clarity	5

Reflection

Structured prompts improved step-by-step reasoning.